From Predictive Coverage to Robust PDE Control: Calibrated Dynamics Ambiguity Sets for Neural Operators

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Abstract

Neural PDE world models can be accurate in an average field norm yet err along the directions that determine a control action. Predictive uncertainty becomes useful for control only when its geometry can be queried by a planner. We construct function-valued dynamics ambiguity sets from a residual Fourier neural operator (FNO) and a replica trained on Gaussian-perturbed labels. Split conformal calibration turns their smoothed disagreement into an isotropic L^2 tube, a normalized ellipsoid, or a simultaneous coordinate box. Robust model predictive control queries these sets through exact adjoint support functions or projected-gradient adversarial rollouts. The analysis separates fixed-grid marginal coverage from deterministic rollout and value bounds that require a uniform one-step error premise. Under compound shift in controlled Burgers dynamics, the audit ellipsoid and simultaneous box both attain 0.906 field coverage, yet their adjoint controllers produce empirical $p90$ costs of 1.0201 and 0.9586. The latter is 10.84% below nominal MPC across 24 matched actuator-gain cases. On a two-dimensional Navier–Stokes benchmark, simultaneous coverage is 0.883 at a nominal 0.90 level, but the bands remain wide because the perturbation scale weakly localizes error. Calibration determines predictive validity; ambiguity geometry and the planner query determine whether that validity becomes useful for control.

Contents

1 Introduction

Neural operators make repeated PDE rollouts cheap enough for online planning. FNOs, DeepONets, and related architectures learn maps between discretized functions and provide fast surrogates for Burgers, Darcy, and Navier–Stokes systems [13] and [14]. Their mathematical foundations are developed in [11] and [4]. A controller, however, never acts on field RMSE. It acts on an objective through the learned rollout. A small error aligned with that objective can change the selected action, while a larger error elsewhere can be irrelevant.

This mismatch persists after uncertainty quantification. Conformal prediction (CP) can provide distribution-free finite-sample coverage for a chosen score under exchangeability [1]; see also [26]. Coverage answers whether a random future field lies in a set. It does not say how the set changes an optimizer. Two sets can therefore attain the same coverage while presenting different worst-case costs to model predictive control (MPC). A control interface must expose the set along objective-sensitive directions and must also specify how one-step uncertainty is lifted through a counterfactual trajectory.

We develop that interface through the chain

$$(\widehat{G}, \widetilde{G}) \longrightarrow \sigma(x,a) \xrightarrow{\text{split CP}} \mathcal{U}_g(x,a) \xrightarrow{\text{support or adversary}} \text{robust MPC}.$$

Sensitivity to label perturbations supplies the local scale σ; a disjoint deployment audit calibrates the geometry g; and the finite-horizon objective queries the resulting field set. The FNO is a replaceable world model, not the architectural contribution. The current planner uses a stagewise trajectory model whose scope remains separate from one-step conformal coverage.

1.1 Related work

Operator learning and uncertainty. FNOs parameterize global integral kernels in Fourier space [13], while the broader neural-operator framework formalizes discretization-invariant maps between function spaces [11]. DeepONet provides a complementary branch–trunk construction [14]. Recent UQ work includes deep ensembles under PDE shift [20], probabilistic neural operators [5], approximate Bayesian neural operators [16], and linearized function-valued Gaussian processes [17]. These methods estimate structured uncertainty, but Bayesian or ensemble dispersion is not automatically a finite-sample coverage statement.

Conformal operator-learning methods address validity in different senses. Risk-controlling quantile neural operators calibrate functional coverage [15]; Conformalized-DeepONet wraps a DeepONet with distribution-free bands [21]; and split CP has been formulated directly in function space [19]. Most closely, Wang et al. train clean-label and perturbed-label FNOs and calibrate a max-type score for simultaneous Navier–Stokes fields in a data-scarce regime [27]. We adopt the perturbation scale as a sample-efficient starting point and extend it with calibrated function-space geometries that can be queried directly by a finite-horizon control objective.

Conformal prediction for robust decisions and control. Conformal uncertainty sets have been connected to robust optimization [10]. In control, Chee et al. combine receding-horizon weighted CP, dynamic constraint tightening, and a predictive reference generator [6]; adaptive CP has also been used in motion planning [8]. SODA-MPC uses conformalized ensembles and runtime OOD adaptation [7], while conformalized system-level synthesis constructs OOD reachable tubes [23]. Those works target finite-dimensional state tubes, constraints, or online drift. Our setting instead starts with a discretized PDE field, compares multiple function-space geometries at a matched audit budget, and uses the finite-horizon objective itself to query the set. We do not inherit their sequential, feasibility, or safety guarantees.

Decision-aware model error. Value-aware model learning replaces uniform prediction quality with error measured through downstream value [9]; calibrated model-based reinforcement learning likewise shows that uncertainty quality can affect planning [18]. Simulation lemmas and Lipschitz analyses propagate one-step model error to long-horizon value error [2]. Distributionally robust model-based reinforcement learning optimizes against dynamics ambiguity in large state spaces [24]. Our contribution is an explicit bridge among these themes: conformal calibration constructs the field set, its support function connects geometry to value sensitivity, and independent closed-loop tests decide whether the construction is useful.

Table 1: Positioning relative to the closest lines of work. "Field" means a joint object over a discretized PDE output; guarantee scopes are not directly comparable.

Work	Uncertainty object	Decision interface	Main guarantee scope
Ma et al. [15]	Functional quantile region	None	Risk-controlled functional coverage
Wang et al. [27]	Perturbation-scaled field box	None	Simultaneous discretized-field coverage
Chee et al. [6]	Weighted-CP state intervals	Constraint tightening and reference generation	Time-varying model accuracy and robust constraints
Srinivasan et al. [23]	Covariance-shaped OOD tube	System-level robust MPC	Weighted-CP OOD coverage and tube robustness
This work	Audit-calibrated L^2 ball, ellipsoid, and field box	Adjoint support or nonlinear adversarial MPC	One-step/behavior-rollout CP; separate deterministic value bound

1.2 Contributions

The paper makes four scoped contributions.

(i) A disjoint audit protocol converts perturbation sensitivity into function-space dynamics sets. Clean and perturbed FNOs use the full proper-training input set, while no calibration label trains either predictor.

(ii) An objective-sensitive query converts those sets into decisions. We derive exact support functions for normalized L^2 ellipsoids and coordinate boxes, embed them in an adjoint robust counterpart, and compare this approximation with nonlinear adversarial rollouts.

(iii) A guarantee ledger separates finite-sample marginal coverage, deterministic multi-step and value propagation under uniform error, and a conditional curvature remainder. None is presented as a substitute for another.

(iv) Matched ablations separate calibration source, ambiguity geometry, and the robust inner query. Controlled Burgers supplies conditional closed-loop evidence, while Navier–Stokes exposes a high-dimensional localization failure. Negative results are retained as evidence about the interface, rather than removed as unsuccessful variants.

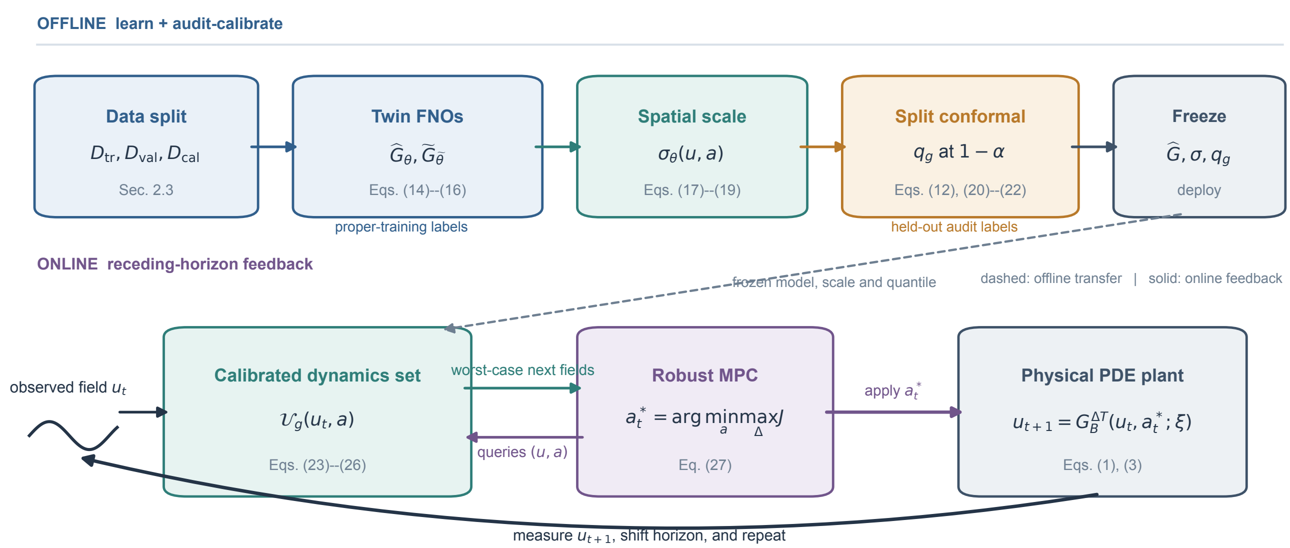

Figure 1: Closed-loop uncertainty-to-control architecture. Offline, the twin FNOs and perturbation scale in Equations (14)–(19) are frozen and the audit split supplies the finite-sample multipliers in Equations (12) and (20)–(22). Online, these objects define the field-valued sets in Equations (23)–(26), which robust MPC queries through Equation (27). The selected action is applied to the physical PDE in Equations (1) and (3); observing the next field closes the receding-horizon loop. Dashed arrows denote offline transfer, not a claim that one-step marginal conformal coverage certifies the counterfactual closed loop.

2 Preliminaries and problem setting

2.1 PDE dynamics and solution operators

2.1.1 Controlled viscous Burgers equation

The control benchmark is the one-dimensional viscous Burgers equation on $\Omega_B = (0, 1)$,

$$\partial_t u + u\partial_x u = \nu\partial_{xx} u + g_{\text{act}} a_t b(x), \qquad u(t, 0) = b_L, \quad u(t, 1) = b_R, \tag{1}$$

where $\nu > 0$ is the viscosity and $a_t \in \mathcal{A} = [-2, 2]$ is held constant on $[t, t + \Delta T)$ with $\Delta T = 0.02$. The scalar g_{act} is the physical actuator gain. The localized actuator shape is parameterized by a center x_c and width s_b,

$$b(x; x_c, s_b) = \frac{\exp[-(x - x_c)^2/(2s_b^2)]}{\max_{z\in[0,1]} \exp[-(z - x_c)^2/(2s_b^2)]}, \qquad x_c = 0.68, \quad s_b = 0.12. \tag{2}$$

Because $x_c \in [0, 1]$, the maximum is attained at $z = x_c$ and the denominator equals one analytically; it is retained to make the unit-peak convention explicit. In the finite-difference implementation, $b_j = b(x_j)$ at interior grid points, the two boundary entries are projected to $b_1 = b_n = 0$, and the interior vector is renormalized to $\max_j b_j = 1$. Thus the endpoints are a discrete boundary convention rather than zeros of the Gaussian itself.

The values $x_c = 0.68$ and $s_b = 0.12$ are fixed synthetic-benchmark design constants, chosen before any data are generated or models are trained. The center lies close to the $x = 0.72$ peak of the state-cost weight in Equation (38); the one-standard-deviation footprint has width $2s_b = 0.24$, preventing nearly pointwise actuation. The normalization leaves $a_t g_{\text{act}}$ responsible for the forcing amplitude. These constants are not statistical estimates and therefore have no associated estimation error in the present experiment. The present deployment shift instead varies the unobserved gain g_{act}. If actuator location or width were uncertain in an application, (x_c, s_b) would have to enter the latent parameter vector and the audit distribution; the guarantees reported here are conditional on the fixed actuator geometry.

For boundary lifting $\ell_b(x) = (1 - x)b_L + xb_R$, define the affine state space $\mathcal{X}_B(b_L, b_R) = \ell_b + H_0^1(0, 1)$. Let $\Xi_B \subset (0, \infty) \times \mathbb{R}^2 \times \mathbb{R}$ denote the admissible parameter set for $\xi_B = (\nu, b_L, b_R, g_{\text{act}})$. The time-$\Delta T$ solution operator is the parameterized family

$$\mathcal{G}_B^{\Delta T}(\cdot, \cdot; \xi_B) : \mathcal{X}_B(b_L, b_R) \times \mathcal{A} \longrightarrow \mathcal{X}_B(b_L, b_R), \qquad (u_t, a_t) \longmapsto u_{t+\Delta T}. \tag{3}$$

The world model observes $\xi_{B,\text{obs}} = (\nu, b_L, b_R)$ but not g_{act}. Varying this latent gain at deployment creates a controlled model shift.

2.1.2 Two-dimensional incompressible Navier–Stokes equation

The field-UQ stress test uses the Reynolds-500 NeuralOperator benchmark [22]. Its spatial domain is the periodic torus $\mathbb{T}^2 = (0, 2\pi)^2$, and its time horizon is $T = 0.5$. In vorticity form, the dynamics are

$$\partial_t\omega + v \cdot \nabla\omega = \nu\Delta\omega - 4\cos(4x_2), \qquad \nabla \cdot v = 0, \qquad \omega(x, 0) = \omega_0(x), \quad (x, t) \in \mathbb{T}^2 \times (0, 0.5]. \tag{4}$$

Here $\omega = \partial_{x_1} v_2 - \partial_{x_2} v_1$ and $v = \nabla^\perp\psi$, where the zero-mean stream function solves $-\Delta\psi = \omega$. The viscosity is fixed at $\nu = \mathrm{Re}^{-1} = 2 \times 10^{-3}$. The archive therefore defines the flow map

$$\mathcal{G}_{\text{NS}}^T : L_0^2(\mathbb{T}^2) \longrightarrow L_0^2(\mathbb{T}^2), \qquad \omega_0 \longmapsto \omega(\cdot, T), \tag{5}$$

where

$$L_0^2(\mathbb{T}^2) = \left\{\omega \in L^2(\mathbb{T}^2) : \int_{\mathbb{T}^2} \omega(x)\, dx = 0\right\} \tag{6}$$

is the space of mean-zero square-integrable periodic vorticity fields. The public input–output pairs contain no action channel. This operator is therefore used only to test high-dimensional field calibration.

2.2 Finite-grid operator-learning maps

Let P_n denote evaluation on a fixed spatial grid. Let I_n denote a boundary-compatible reconstruction for Burgers and a periodic reconstruction for Navier–Stokes. For Burgers, $n = 64$ and the ground-truth one-step transition is

$$\begin{aligned} G_{B,n} &: \mathbb{R}^{64} \times [-2, 2] \times \Xi_B \longrightarrow \mathbb{R}^{64}, \\ &G_{B,n}(x, a, \xi_B) = P_n\mathcal{G}_B^{\Delta T}(I_n x, a; \xi_B). \end{aligned} \tag{7}$$

The residual FNO used by the controller approximates the partially observed map

$$\widehat{G}_{\theta,B} : \mathbb{R}^{64} \times [-2,2] \times \Xi_{B,\mathrm{obs}} \longrightarrow \mathbb{R}^{64}. \tag{8}$$

Here $\Xi_{B,\mathrm{obs}}$ contains the observed triples (ν, b_L, b_R). Its training target is $y = G_{B,n}(x, a, (\xi_{B,\mathrm{obs}}, g_{\mathrm{act}}))$ even though the gain is not an input. The learned operator consequently represents the source-regime input–output relation rather than a gain-conditioned physical solver.

For Navier–Stokes, vectorizing the 128×128 fields gives $n = 16384$ and

$$\begin{aligned} G_{\mathrm{NS},n} &: \mathbb{R}^{16384} \longrightarrow \mathbb{R}^{16384}, \\ G_{\mathrm{NS},n}(x) &= P_n \mathcal{G}_{\mathrm{NS}}^T(I_n x). \end{aligned} \tag{9}$$

The corresponding FNO has the same Euclidean input and output spaces, $\widehat{G}_{\theta,\mathrm{NS}} : \mathbb{R}^{16384} \to \mathbb{R}^{16384}$, and approximates $G_{\mathrm{NS},n}$ on the distribution of archived vorticity fields. We use

$$\langle v, w \rangle_n = \frac{1}{n} \sum_{j=1}^{n} v_j w_j, \qquad \|v\|_{2,n}^2 = \langle v, v \rangle_n \,. \tag{10}$$

Nonuniform quadrature weights can replace $1/n$. A "function-space set" in the experiments is a joint set for the full discretized field. Fixed-grid coverage is not a continuum or resolution-transfer guarantee.

2.3 Control objective, data roles, and deployment shift

For the Burgers controller, the finite-horizon objective is

$$J_H(x_0, a_{0:H-1}) = \sum_{t=1}^{H} \ell(x_t) + \sum_{t=0}^{H-1} r(a_t), \tag{11}$$

where lower cost is preferred. The learned map in Equation (8) is rolled out inside MPC and is therefore queried at optimizer-induced states and actions.

We distinguish four data roles:

$$\mathcal{D}_{\mathrm{tr}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{tr}}}, \qquad \mathcal{D}_{\mathrm{val}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{val}}},$$
$$\mathcal{D}_{\mathrm{cal}} = \{(x_i, a_i, y_i)\}_{i=1}^{m}, \qquad \mathcal{D}_{\mathrm{te}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{te}}}.$$

Proper training and validation come from a source regime. The deployment audit $\mathcal{D}_{\mathrm{cal}}$ is a small labeled sample from the target regime and is used only to compute conformal order statistics. The test set remains untouched until final evaluation. The clean and perturbed models, smoothing kernel, and floor rule are measurable with respect to proper-training randomness and are fixed before $\mathcal{D}_{\mathrm{cal}}$ is scored.

This is an *audit-calibrated* setting: validity requires that calibration and target examples are exchangeable after conditioning on the trained models. If the deployment distribution changes again after the audit, ordinary split CP does not protect against that new shift. Weighted or adaptive CP [3], or covariate-shift calibration [25], is a possible extension, not a guarantee claimed here.

2.4 Split conformal prediction on the fixed grid

Let $Z_i = (X_i, A_i, Y_i)$ and let $S(Z_i)$ be a fixed scalar nonconformity score. With calibration scores $S_{(1)} \le \cdots \le S_{(m)}$, target miscoverage $\alpha \in (0,1)$, and

$$k = \lceil (m+1)(1-\alpha) \rceil \,, \qquad q_{1-\alpha} = S_{(k)}, \tag{12}$$

the standard convention sets $q_{1-\alpha} = +\infty$ if $k > m$. Equivalently, the empirical distribution is augmented by an atom at $+\infty$. For a new input (x, a), the prediction set is

$$C(x, a) = \{y : S(x, a, y) \le q_{1-\alpha}\}. \tag{13}$$

Under exchangeability, the rank of the new score among the $m+1$ scores yields marginal coverage at least $1 - \alpha$. The word *marginal* matters: it is not conditional coverage at every state–action pair and it is not automatic simultaneous coverage over a counterfactual MPC rollout.

3 Methodology

3.1 Control-oriented dynamics ambiguity sets

The method maps a predictive scale to a state–action-dependent dynamics set $\mathcal{U}_\theta(x,a) \subset \mathbb{R}^n$ and exposes its geometry to a robust planner. We evaluate three properties separately. Predictive validity concerns the predeclared conformal containment event. Statistical efficiency concerns set width or volume. Control utility concerns paired mean cost, upper-tail cost, and control effort on independent closed-loop cases. A set can satisfy the first property without improving the third, so coverage is supporting evidence rather than a definition of control quality.

3.2 Residual FNO world model

The base one-step world model is a residual FNO,

$$\widehat{G}_\theta(x,a,\xi_{\text{obs}}) = x + \delta_\theta(x,a,\xi_{\text{obs}}). \tag{14}$$

For Burgers, the five input channels are the current field, the action multiplied by a fixed actuator profile, viscosity, a linear extension of the two boundary values, and the spatial coordinate. Each block applies a spectral convolution and a pointwise linear map inside a ResNet-style skip connection. Writing v_ℓ for a hidden field, one block has the form

$$v_{\ell+1} = v_\ell + \phi\big(W_\ell v_\ell + \mathcal{F}^{-1}(R_\ell \odot \mathcal{F} v_\ell)\big), \tag{15}$$

where R_ℓ retains the lowest Fourier modes and W_ℓ is a learned pointwise map. The FNO therefore combines global spectral mixing, a local linear path, and a residual connection. The final boundary coordinates are projected exactly to the supplied Dirichlet values. The full run uses width 32, 14 Fourier modes, four residual blocks, and 240,454 trainable parameters across the clean and perturbed operators.

FNO is chosen because it is a standard field propagator [13]; the broader framework is reviewed in [11]. Nothing in the calibration or robust-control construction requires Fourier layers. Replacing it by another differentiable operator changes the mean and scale quality, not the logic of the method.

3.3 Perturbation-based uncertainty scale

Following the stability principle used for data-scarce operator UQ [27], we train a second operator with the same architecture and the same proper-training inputs but perturbed labels:

$$\widetilde{y}_i = y_i + \eta_i, \qquad \eta_i \overset{\text{iid}}{\sim} \mathcal{N}(0, c^2 s_y^2 I_n). \tag{16}$$

Let $\widetilde{G}_{\widetilde{\theta}}$ be its prediction. The raw disagreement, smoothed disagreement, and stabilized scale are

$$d(x,a) = \left|\widehat{G}_\theta(x,a) - \widetilde{G}_{\widetilde{\theta}}(x,a)\right|, \tag{17}$$

$$\bar{d}(x,a) = K * d(x,a), \tag{18}$$

$$\sigma(x,a) = \bar{d}(x,a) \vee \tau_0. \tag{19}$$

The convolution is an odd moving-average window (five coordinates for Burgers; 15×15 for Navier–Stokes). On Burgers, τ_0 is 10% of the median interior disagreement on $\mathcal{D}_{\text{tr}}$, lower bounded by 10^{-6}. Smoothing prevents isolated near-zero denominators from dominating a max-type score; the floor is fixed before calibration. The raw scale has no coverage interpretation. Split conformal calibration supplies validity.

3.4 Perturbation-based nonconformity scores and calibration

For residual $r = y - \widehat{G}_\theta(x,a)$, we compare three geometries.

$$S_{\text{L2}}(x,a,y) = \frac{\|r\|_{2,n}}{\|\sigma(x,a)\|_{2,n}}, \tag{20}$$

$$S_2(x,a,y) = \left\|r \odot \sigma(x,a)^{-1}\right\|_{2,n}, \tag{21}$$

$$S_\infty(x,a,y) = \left\|r \odot \sigma(x,a)^{-1}\right\|_\infty. \tag{22}$$

Here $\odot$ denotes coordinate-wise (Hadamard) multiplication and $\sigma(x,a)^{-1}$ denotes the vector of coordinate-wise reciprocals; the floor $\tau_0 > 0$ in Equation (19) makes every reciprocal finite. Applying Equation (12) separately gives q_{L2}, q_2, and q_∞. The corresponding one-step dynamics sets are

$$\mathcal{U}_{\mathrm{L2}}(x,a) = \left\{\widehat{G}_\theta(x,a) + \Delta : \|\Delta\|_{2,n} \le q_{\mathrm{L2}}\,\|\sigma(x,a)\|_{2,n}\right\}, \tag{23}$$

$$\mathcal{U}_2(x,a) = \left\{\widehat{G}_\theta(x,a) + \Delta : \left\|\Delta \odot \sigma(x,a)^{-1}\right\|_{2,n} \le q_2\right\}, \tag{24}$$

$$\mathcal{U}_\infty(x,a) = \left\{\widehat{G}_\theta(x,a) + \Delta : |\Delta_j| \le q_\infty \sigma_j(x,a),\ j = 1,\dots,n\right\}. \tag{25}$$

The first is an isotropic L^2 tube baseline. The second preserves heteroscedastic anisotropy but covers an L^2 event. The third targets simultaneous containment of every output coordinate. The box is usually wider because one failure anywhere in the field breaks the event.

3.5 Practical implementation

Algorithm 1 summarizes training, calibration, and deployment. The geometry index $g \in \{\mathrm{L2}, 2, \infty\}$ selects one of Equations (20) to (22) and its associated set.

Algorithm 1 Perturbation-calibrated ambiguity sets for robust PDE control

Require: Disjoint $\mathcal{D}_{\mathrm{tr}}, \mathcal{D}_{\mathrm{val}}, \mathcal{D}_{\mathrm{cal}}, \mathcal{D}_{\mathrm{te}}$; miscoverage α; noise scale c; smoother K; geometry g
Ensure: Frozen operators, multiplier q_g, and deployed set $\mathcal{U}_g$

- $\theta \leftarrow \textsc{FitFNO}(\mathcal{D}_{\mathrm{tr}}$ with clean targets$, \mathcal{D}_{\mathrm{val}})$
- Form $\widetilde{y}_i = y_i + \eta_i$ with $\eta_i \sim \mathcal{N}(0, c^2 s_y^2 I)$ on $\mathcal{D}_{\mathrm{tr}}$
- $\widetilde{\theta} \leftarrow \textsc{FitFNO}(\mathcal{D}_{\mathrm{tr}}$ with $\widetilde{y}_i, \mathcal{D}_{\mathrm{val}})$
- Freeze $\widehat{G}_\theta$ and $\widetilde{G}_{\widetilde{\theta}}$
- Fix K and τ_0 using proper-training predictions only
- **for** $(x_i, a_i, y_i) \in \mathcal{D}_{\mathrm{cal}}$ **do**
  - $\mu_i \leftarrow \widehat{G}_\theta(x_i, a_i)$, $\widetilde{\mu}_i \leftarrow \widetilde{G}_{\widetilde{\theta}}(x_i, a_i)$
  - $\sigma_i \leftarrow K * |\mu_i - \widetilde{\mu}_i| \vee \tau_0$
  - $r_i \leftarrow y_i - \mu_i$; compute $S_i^{(g)}$ from Equations (20) to (22)
- **end for**
- $k \leftarrow \lceil (|\mathcal{D}_{\mathrm{cal}}| + 1)(1-\alpha) \rceil$; $q_g \leftarrow S_{(k)}^{(g)}$
- Store weights, K, τ_0, q_g, split metadata, and seed

Deployment:

- **for** each MPC query (x_t, a) **do**
  - Evaluate $\widehat{G}_\theta(x_t, a)$ and $\sigma(x_t, a)$
  - Construct $\mathcal{U}_g(x_t, a)$ and solve the robust inner problem
- **end for**
- Evaluate coverage and control once on $\mathcal{D}_{\mathrm{te}}$

Checkpoint selection uses only $\mathcal{D}_{\mathrm{val}}$. The smoothing rule, floor, and both operators are fixed before calibration, and no label in $\mathcal{D}_{\mathrm{cal}}$ is reused for final coverage or closed-loop evaluation.

3.6 From one-step calibrated sets to multi-step robust control

Split conformal calibration constructs a set for one random transition, while MPC optimizes a counterfactual trajectory. The lifting from local prediction sets to trajectory uncertainty is therefore a separate modeling step. For a candidate action sequence, define the admissible perturbations recursively as

$$\begin{aligned}\mathfrak{E}_H(x_0, a_{0:H-1}) = \{\Delta_{0:H-1} : \Delta_t \in \mathcal{E}(x_t, a_t),& \\ x_{t+1} = \widehat{G}_\theta(x_t, a_t) + \Delta_t,& \quad 0 \le t < H\},\end{aligned} \tag{26}$$

where $\mathcal{E}(x_t, a_t)$ is the centered ellipsoid or box from Equations (24) and (25). The robust planning problem is

$$\min_{a_{0:H-1} \in \mathcal{A}^H} \max_{\Delta_{0:H-1} \in \mathfrak{E}_H(x_0, a_{0:H-1})} \left\{\sum_{t=1}^{H} \ell(x_t) + \sum_{t=0}^{H-1} r(a_t)\right\}. \tag{27}$$

Because $\mathcal{E}(x_t, a_t)$ changes with the perturbed state, $\mathfrak{E}_H$ is not generally a fixed Cartesian product. The implementation allows a separate disturbance choice at each predicted state but imposes no explicit temporal coupling beyond the state recursion. It is therefore a stagewise adversarial approximation, sometimes described as time-rectangular after conditioning on the current state. This computational choice is not an independence claim about physical model error, and one-step marginal coverage does not provide joint coverage of Equation (26).

The remaining methodological problem is to construct trajectory-level sets that preserve temporal coherence and remain valid under policy-induced state and action distributions. Candidate extensions include a shared latent operator perturbation across the horizon, low-rank spatiotemporal sets, and trajectory conformal scores calibrated under a declared behavior or target policy. Their value should be assessed at matched trajectory coverage by set tightness and closed-loop cost, not by coverage alone.

3.6.1 Exact support and the adjoint robust counterpart

For the first-order robust counterpart, we freeze the local centered sets on the nominal rollout: $\mathcal{E}_t^0 = \mathcal{E}(\widehat{x}_t, a_t)$ and $\mathcal{P}_H^0 = \prod_{t=0}^{H-1} \mathcal{E}_t^0$. This frozen approximation is a Cartesian product, so its support separates across time.

Let λ_{t+1} denote the full derivative of the remaining nominal horizon cost with respect to x_{t+1}, expressed in the normalized inner product: $\delta J = \langle \lambda_{t+1}, \Delta_t \rangle_n$. Its value includes the effect of Δ_t on every later predicted state. The support of a centered set $\mathcal{E}$ is $h_{\mathcal{E}}(\lambda) = \sup_{\Delta \in \mathcal{E}} \langle \lambda, \Delta \rangle_n$.

Proposition 1 (Support functions of the calibrated geometries)**.** *For positive scale σ,*

$$h_{\mathcal{E}_2}(\lambda) = q_2 \left\| \lambda \odot \sigma \right\|_{2,n}, \tag{28}$$

$$h_{\mathcal{E}_\infty}(\lambda) = \frac{q_\infty}{n} \sum_{j=1}^{n} |\lambda_j| \sigma_j. \tag{29}$$

Proof. For the ellipsoid, set $\Delta = \sigma \odot z$ and apply Cauchy–Schwarz under $\langle \cdot, \cdot \rangle_n$; equality is attained by choosing z parallel to $\lambda \odot \sigma$. For the box, every coordinate is independent, and the maximizer is $\Delta_j = q_\infty \sigma_j \operatorname{sign}(\lambda_j)$. $\square$

Linearizing the stacked inner problem over $\mathcal{P}_H^0$ gives

$$\widehat{J}_H(a_{0:H-1}) + \sum_{t=0}^{H-1} h_{\mathcal{E}_t^0}(\lambda_{t+1}). \tag{30}$$

This formula is the decision interface: two sets can have similar coverage but produce different robust costs because their support differs along λ_{t+1}. Automatic differentiation supplies all finite-horizon adjoints in one reverse pass.

3.6.2 Nonlinear adversarial rollout

The first-order objective can miss state-dependent scales and nonlinear error propagation. Our second inner solver parameterizes $\Delta_t = \sigma(x_t, a_t) \odot z_t$, initializes $z_{0:H-1} = 0$, performs projected gradient ascent on the full rollout cost, and projects each z_t onto either $\|z_t\|_{2,n} \le q_2$ or $\|z_t\|_\infty \le q_\infty$. For Burgers, $H = 8$ and $n = 64$, so the inner variable has $Hn = 512$ coordinates for each fixed action sequence. The objective is nonconvex because the rollout recursively composes nonlinear FNO blocks and recomputes $\sigma(x_t, a_t)$ at each perturbed state.

The experiment uses one zero initialization, three ascent steps, step size 0.8, and no random restarts. Projection makes the returned perturbation feasible for the stagewise sets. Its rollout cost is therefore a lower bound on the exact inner supremum, not an upper bound, global optimum, or certified optimality gap. We use this solver only as a compute-bounded adversarial-search baseline. The adjoint support and this nonlinear search answer different questions and neither certifies the full nonlinear min–max problem.

3.6.3 Outer optimization and receding-horizon deployment

The outer action sequence is optimized by the cross-entropy method (CEM): 128 candidates, 16 elites, four iterations, horizon eight, and action bound $|a_t| \le 2$. The first action is applied to the numerical plant and the entire problem is replanned at the next observed state. Nominal, isotropic-tube, adjoint-ellipsoid, adjoint-box, and adversarial controllers use the same world model and CEM budget.

3.7 Theoretical guarantees and scope

3.7.1 Finite-sample coverage on a fixed discretization

Theorem 1 (One-step split-conformal coverage)**.** *Condition on $\mathcal{D}_{\mathrm{tr}}$, model-training randomness, perturbation noise, and all scale hyperparameters. If the m calibration triples and one new deployment triple are exchangeable, then each set constructed by its matching score satisfies*

$$\Pr\left\{G_{\star,n}(X,A) \in \mathcal{U}(X,A)\right\} \geq 1-\alpha.$$

For $\mathcal{U}_\infty$, the event contains all n output coordinates simultaneously.

Proof. After conditioning, the score is fixed and identical for every example. The m calibration scores and the new score are therefore exchangeable. The rank argument for Equation (12), including the $+\infty$ convention, gives the result. For the max score, $S_\infty \leq q_\infty$ is equivalent to all n coordinate inequalities in Equation (25). ☐

The theorem concerns one random transition. To obtain a different, valid trajectory statement, predeclare a behavior policy and define on whole length-H trajectories

$$S^{\mathrm{traj}} = \max_{0 \leq t < H} \max_{1 \leq j \leq n} \frac{|x_{t+1,j} - \widehat{x}_{t+1,j}|}{\sigma_j(\widehat{x}_t, a_t)}. \tag{31}$$

Split-conformal calibration of this scalar score on exchangeable trajectories covers all times and coordinates of one new behavior-policy trajectory with probability at least $1-\alpha$. It does not certify counterfactual action sequences chosen by MPC.

3.7.2 First-order robustification and curvature

Let $\boldsymbol{\Delta} = (\Delta_0, \ldots, \Delta_{H-1})$ and let $F(\boldsymbol{\Delta})$ be the finite-horizon cost after injecting the stacked perturbation while the nominal-rollout sets remain frozen.

Proposition 2 (Conditional second-order remainder)**.** *Suppose F is twice differentiable and $\|\nabla^2 F(\boldsymbol{z})\|_{\mathrm{op}} \leq M_H$ on the line segments joining 0 to every $\boldsymbol{\Delta} \in \mathcal{P}_H^0$. Then*

$$\sup_{\boldsymbol{\Delta} \in \mathcal{P}_H^0} F(\boldsymbol{\Delta}) \leq F(0) + \sum_{t=0}^{H-1} h_{\mathcal{E}_t^0}(\lambda_{t+1}) + \frac{M_H}{2} \sup_{\boldsymbol{\Delta} \in \mathcal{P}_H^0} \|\boldsymbol{\Delta}\|_2^2. \tag{32}$$

Proof. Taylor's theorem gives $F(\boldsymbol{\Delta}) \leq F(0) + \langle \nabla F(0), \boldsymbol{\Delta} \rangle + (M_H/2)\|\boldsymbol{\Delta}\|_2^2$. The support of $\mathcal{P}_H^0$ is the sum of its component supports. Taking the supremum gives Equation (32). ☐

The adjoint objective retains only the first two terms. A learned or audit-fitted quadratic feature is not a certified curvature bound unless a uniform M_H is established.

3.7.3 Multi-step propagation under uniform one-step error

Conformal coverage is distributional. The following result uses a different, deterministic assumption.

Assumption 1 (Uniform model error and Lipschitz dynamics)**.** *On a forward-invariant state–action region $\mathcal{K}$,*

$$\left\|G_{\star,n}(x,a) - \widehat{G}_\theta(x,a)\right\|_{2,n} \leq \epsilon, \qquad \|G_{\star,n}(x,a) - G_{\star,n}(x',a)\|_{2,n} \leq L_G \|x - x'\|_{2,n}. \tag{33}$$

Proposition 3 (Open-loop rollout error)**.** *For a common action sequence, equal initial states, and $e_t = \|x_t - \widehat{x}_t\|_{2,n}$,*

$$e_{t+1} \leq L_G e_t + \epsilon, \qquad e_h \leq \epsilon \sum_{k=0}^{h-1} L_G^k. \tag{34}$$

Proof. Add and subtract $G_{\star,n}(\widehat{x}_t, a_t)$, then use the two parts of Equation (33). Iterating the scalar recurrence with $e_0 = 0$ gives the geometric sum. ☐

3.7.4 Discounted value and policy transfer

For a feedback policy π, define the closed-loop map $F_G^\pi(x) = G(x, \pi(x))$ and stage reward $r_\pi(x) = r(x, \pi(x))$. The value is $V_G^\pi(x_0) = \sum_{t=0}^{\infty} \gamma^t r_\pi(x_t)$.

Theorem 2 (Fixed-policy value error)**.** *Suppose the true and learned closed-loop maps for a fixed policy share the state Lipschitz constant L_G, their one-step discrepancy is uniformly at most ϵ, r_π is L_r-Lipschitz, and $\gamma L_G < 1$. Then*

$$\left| V_{G_\star}^\pi(x_0) - V_{\widehat{G}}^\pi(x_0) \right| \le B(\epsilon) := \frac{\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}. \tag{35}$$

Proof. By Equation (34), $e_t \le \epsilon \sum_{k=0}^{t-1} L_G^k$. Hence

$$\sum_{t=1}^{\infty} \gamma^t L_r e_t \le L_r \epsilon \sum_{k=0}^{\infty} L_G^k \sum_{t=k+1}^{\infty} \gamma^t = \frac{\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}.$$

□

Corollary 1 (Policy transfer with optimization error)**.** *Suppose the preceding assumptions hold uniformly over a policy class. Let $\pi^\star$ maximize true-model value, and let $\widehat{\pi}$ satisfy*

$$\sup_\pi V_{\widehat{G}}^\pi(x_0) - V_{\widehat{G}}^{\widehat{\pi}}(x_0) \le \delta_{\text{opt}}. \tag{36}$$

Then

$$V_{G_\star}^{\pi^\star}(x_0) - V_{G_\star}^{\widehat{\pi}}(x_0) \le \frac{2\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)} + \delta_{\text{opt}}. \tag{37}$$

If $L_G \le 1$, the model term is at most $2\gamma L_r \epsilon/(1-\gamma)^2$.

Proof. Insert and subtract the two learned-model values. The two model-transfer terms are bounded by $B(\epsilon)$, and learned-model suboptimality contributes δ_{opt}. □

The CEM controller has finite sampling and four optimization iterations, so δ_{opt} is not known in our experiment. The corollary explains the target scaling; it is not a certificate for the reported CEM policy.

Numerical scaling witness. The deterministic rate can be checked without conformal calibration. For the scalar witness $G(x) = L_G x$, $\widehat{G}(x) = L_G x + \epsilon$, and reward $-L_r|x|$, the fixed-policy bound is attained. Across six values each of ϵ and γ, the log–log slope is 1.000 in ϵ. After normalizing the value gap by $\gamma\epsilon$, the slope against effective horizon $(1-\gamma)^{-1}$ is 2.000, with maximum relative numerical error 2.55×10^{-15}. Thus the calculation verifies the deterministic rate in Equation (37); it does not turn marginal conformal coverage into a uniform model-error guarantee.

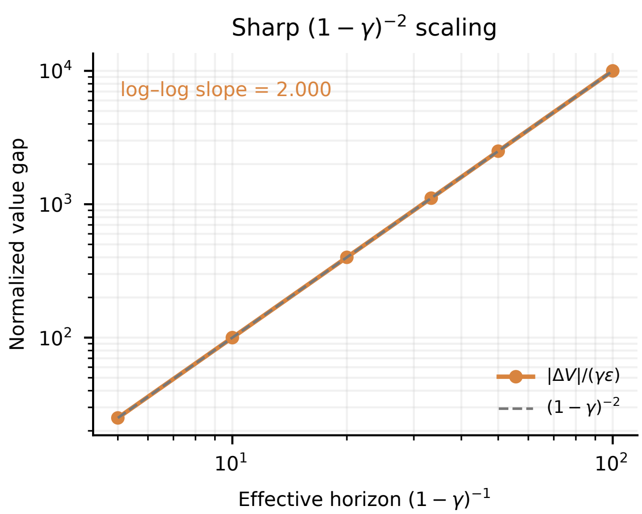

Figure 2: Deterministic scaling audit for the policy-transfer result. Normalizing by $\gamma\epsilon$ exposes the predicted quadratic dependence on the effective discount horizon. The generic policy-transfer bound is twice the fixed-policy witness plus optimization error.

For ten controlled Burgers rollouts with an exactly injected one-step error, the maximum observed gap-to-finite-horizon-bound ratio is 0.359. On the learned FNO's finite visited set, the maximum and mean gap-to-local-envelope ratios are 0.124 and 0.028. These are empirical local audits, not global PDE certificates.

3.7.5 Guarantee separation

Table 2: The three statement types used in the paper. None implies either of the others without additional assumptions.

Statement	Assumption	Event or quantity	What it does not provide
Split CP	Exchangeable audit/test scores	Marginal one-step field containment	Uniform error over a region or MPC safety
Trajectory CP	Exchangeable behavior-policy trajectories	All time/coordinates for one such trajectory	Counterfactual MPC action coverage
Value bound	Deterministic uniform ϵ, Lipschitz maps/reward	Rollout and policy-transfer error	A probability statement or conformal validity
Closed-loop test	Independent matched cases	Empirical mean/tail cost	A theorem beyond the tested design

4 Experiments

The numerical study follows the progression used in operator-learning conformal evaluations: model fit is checked first, predictive validity and set efficiency are then evaluated on held-out fields, and ablations vary one component while fixing the trained operators and test examples. Closed-loop control is reported only for the action-conditioned Burgers system. The Navier–Stokes data are used as a separate high-dimensional calibration test.

4.1 Dataset and experimental protocol

The Burgers plant in Equation (1) is solved on 64 grid points with an upwind convective flux, centered diffusion, solver step 2×10^{-4}, and control interval 0.02. The Gaussian actuator is centered at $x = 0.68$. The stage cost is

$$\ell(u) + r(a) = \frac{1}{n}\sum_{j=1}^{n} w(x_j)u_j^2 + 0.002a^2, \quad w(x) = 0.15 + 0.85\exp\left[-\frac{1}{2}\left(\frac{x-0.72}{0.14}\right)^2\right]. \tag{38}$$

Training and deployment differ in viscosity, boundary values, initial-field amplitude, and latent actuator gain g_{act}. The model observes viscosity and boundaries but not the actuator gain.

Two-dimensional Navier–Stokes benchmark. We additionally use the official NeuralOperator incompressible Navier–Stokes benchmark [22]. The downloaded Reynolds-500 archive supplies 128×128 vorticity input–output fields. A clean FNO and a perturbed-label FNO use the same 800 proper-training inputs and clean or perturbed targets, respectively. These indices are disjoint from validation, audit, and test roles. A max-score conformal audit follows model selection. The dataset contains no action channel; it tests field UQ and dimensionality, not closed-loop control.

Table 3: Disjoint data roles in the final experiments. Burgers one-step test size is per regime. Trajectory calibration/test splits are separate from all one-step splits.

Benchmark	Train	Validation	Source cal.	Deploy. audit	Test	Traj. audit/test
Burgers	4000	600	800	300	800	200/400
Navier–Stokes	800	50	–	200	300	–

4.2 Baselines, metrics, and implementation

All Burgers controllers use the same residual FNO pair. We compare: uncontrolled dynamics; nominal MPC; source-calibrated isotropic L^2 robust MPC; deployment-audit L^2 robust MPC; audit ellipsoid with adjoint support;

audit simultaneous box with adjoint support; and audit ellipsoid with nonlinear adversarial rollouts. This design separates access to deployment labels from ambiguity geometry and from the inner solver.

The predictive-set names encode two independent choices: where the conformal multiplier is calibrated and how the field residual is measured. All variants share the frozen FNO pair, the scale $\sigma(x, a)$, the target coverage $1 - \alpha = 0.90$, and the same 800 compound-shift test transitions. *Source* L^2 uses the 800 source-regime calibration transitions with S_{L2} and the isotropic tube $\mathcal{U}_{\mathrm{L2}}$ in Equations (20) and (23); it tests transfer under deployment shift. *Audit* L^2 uses the same score and set but re-estimates the multiplier on 300 target-regime audit transitions, thereby isolating the effect of calibration population. *Audit ellipsoid* uses that deployment audit with S_2 and $\mathcal{U}_2$ in Equations (21) and (24), combining coordinate-wise scales through one normalized L^2 score. *Audit simultaneous box* uses the same audit labels with S_∞ and $\mathcal{U}_\infty$ in Equations (22) and (25); its max score requires all grid coordinates to be covered in the same test field. Thus Source L^2 versus Audit L^2 isolates calibration shift, while the three audit rows isolate ambiguity geometry.

We report realized function-set coverage, decision-linear coverage (residual projected onto a cost direction), mean support, radius–error association, behavior-trajectory coverage, paired mean control cost, empirical $p90$ cost, and mean absolute action. Twenty-four combined-shift actuator-gain cases are matched across controllers, each with 20 receding-horizon steps. These cases share the same initial field and physical parameters except actuator gain; they are a controlled sweep rather than 24 independent PDE draws.

The two FNOs are trained for 60 epochs with AdamW, batch size 64, learning rate 2×10^{-3}, and weight decay 10^{-5}. The label-noise multiplier is $c = 0.05$. For every CEM candidate, the nonlinear adversary uses the single-start three-step projected-gradient budget specified above. Thus its reported row is an end-to-end controller result under a fixed search budget, not an evaluation of the exact inner maximum. The released package contains code, split rules, source JSON/CSV files, a portable notebook, and plotting scripts. Large raw data and full-run checkpoints are regenerated locally and are not stored in version control.

4.3 Ablation protocol and within-group controls

Following the controlled-comparison style of Wang et al. [27], every ablation group uses identical evaluation examples across variants and changes only the named interface. All reported groups condition on the seed-27 FNO pair; repeated cases quantify conditional variation, not training-seed variation. Table 4 records the comparison contract.

Table 4: Within-group ablation settings. “Fixed” lists the quantities shared by every row in the group; the final column is the only intended change.

Group	Fixed within the group	Varied factor
A: calibration and geometry	FNO pair, disagreement scale, 90% target, 800 ordered compound-shift transitions	Source versus deployment audit; L^2 tube versus ellipsoid versus coordinate box
B: robust-control interface	FNO pair, 24 matched plants, common initial field and physical parameters, 20 steps, common CEM seeds	Nominal, isotropic penalty, adjoint support, or single-start three-step PGD adversarial rollout
C: value-bound construction	Same 60 calibration and 160 test trajectories, observed gaps, horizon 20, $\gamma = 0.95$, 90% value target	Global recursion, local recursion, adjoint support, or adjoint plus curvature

Within Group A, source-L^2 versus audit-L^2 isolates calibration access; the three audit-calibrated rows isolate geometry. Within Group B, the two adjoint rows isolate geometry at a common first-order interface, whereas the ellipsoid-adjoint and adversarial-ellipsoid rows isolate the inner query at a fixed set. Within Group C, each raw bound is separately scaled on the common calibration trajectories before evaluation on the same independent test trajectories. We do not import the smoothing or label-noise ablation numbers from Wang et al.; those require retraining and are not claimed as our data.

4.4 World-model fit and uncertainty localization

The base FNO ends with training and validation MSEs 1.01×10^{-4} and 7.67×10^{-5}; the perturbed replica has validation MSE 9.87×10^{-5}. Under combined shift, the mean field L^2 error is 0.00780, and the perturbation scale has a modest radius–error correlation of 0.275. The comparable validation losses show that label perturbation preserves the dominant one-step dynamics. The modest correlation, however, means that disagreement only weakly ranks where the base FNO will err. Conformal calibration can correct event frequency, but it cannot create coordinate-wise localization that is absent from the scale. An accurate mean predictor can therefore coexist with a wide calibrated set.

4.5 Ablation A: audit calibration restores coverage under shift

The first contrast in Table 5 changes only the calibration population. Source L^2 covers 593 of 800 compound-shift fields (0.741), 15.9 percentage points below the 0.90 target. Re-estimating the multiplier on the deployment audit, without retraining either FNO, raises coverage to 707 of 800 (0.884). This paired change identifies calibration mismatch, rather than a change in model architecture, as the source of the largest coverage loss.

The second contrast changes only set geometry on the common deployment audit. The ellipsoid and simultaneous box each cover 725 of 800 fields (0.906), but they are not equivalent decision objects. The box attains 1.000 decision-linear coverage with mean support 0.04422, whereas the ellipsoid attains 0.986 with support 0.02012. The max-coordinate event protects every grid value and pays for directions that the current cost may not emphasize. Equal field coverage can therefore conceal more than a twofold difference in the uncertainty presented to the controller.

Table 5: Combined-shift one-step results. The target coverage is 0.90. Support is the half-width in the evaluated decision direction.

Set	Field coverage	Mean radius	Decision coverage	Mean support
Source L^2	0.741	0.01141	0.820	0.00650
Audit L^2	0.884	0.01882	0.934	0.01072
Audit ellipsoid	0.906	0.03242	0.986	0.02012
Audit simultaneous box	0.906	0.22729	1.000	0.04422

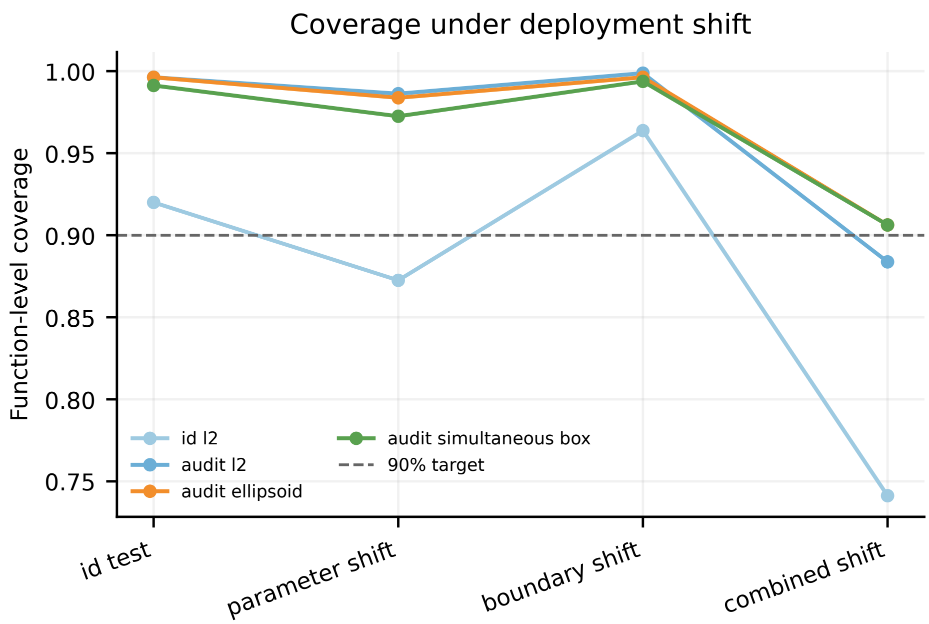

Figure 3: Realized one-step coverage across in-distribution, viscosity-shift, boundary-shift, and combined-shift test sets. Source calibration degrades most under combined shift; deployment-audit geometries recover the nominal region at different widths. Each regime contains 800 test transitions.

The separately calibrated behavior-trajectory max score attains empirical coverage 0.88 on 400 held-out length-10 trajectories at target 0.90. We report the observed proportion rather than rounding it to the target. Its scope is the fixed behavior-policy distribution in Equation (31). For scale, the audit-L^2, ellipsoid, box, and trajectory events correspond to 707/800, 725/800, 725/800, and 352/400 covered examples. Their descriptive 95% Wilson intervals all contain 0.90. These intervals are an empirical stability check, not the source of the conformal guarantee.

4.6 Ablation B: matched-coverage sets yield different tail costs

The matched control sweep exposes the consequence of ambiguity geometry. The audit ellipsoid and simultaneous box have identical realized field coverage of 0.906, yet their adjoint controllers reach empirical $p90$ costs of 1.0201 and 0.9586. Relative to nominal MPC, the box lowers mean cost by 1.89% and $p90$ by 10.84%; audit L^2 lowers them by 1.92% and 7.79%. All robust variants improve the empirical tail, while every mean improvement remains below 2%. Robustification therefore changes the difficult actuator-gain cases more than the center of the matched sweep.

The box-adjoint controller also reduces mean $|a|$ from 1.278 to 1.101. This association is consistent with less aggressive actions under latent gain uncertainty, but the experiment does not identify action moderation as the sole cause of the cost reduction. The nonlinear adversarial ellipsoid is not best under the single-start three-step PGD and fixed CEM budgets. This outcome does not establish that nonlinear min–max robustification is inferior. It may instead reflect local optimization error in the high-dimensional nonconvex inner search. The comparison therefore concerns the implemented query mechanisms under stated budgets, not their globally optimized counterparts. Because the 24 cases form one matched gain sweep, these percentages are conditional effect sizes rather than population-level estimates across training seeds.

Table 6: Closed-loop cost over 24 matched actuator-gain cases. Changes are relative to nominal MPC; lower is better.

Controller	Mean	Mean change	$p90$	$p90$ change	Mean $|a|$
Nominal MPC	0.5797	–	1.0752	–	1.278
Source-L^2 robust	0.5724	-1.27%	1.0250	-4.66%	1.185
Audit-L^2 robust	0.5686	-1.92%	0.9914	-7.79%	1.115
Ellipsoid-adjoint	0.5747	-0.88%	1.0201	-5.12%	1.230
Box-adjoint	0.5688	-1.89%	0.9586	-10.84%	1.101
Adversarial ellipsoid	0.5764	-0.58%	1.0532	-2.04%	1.262

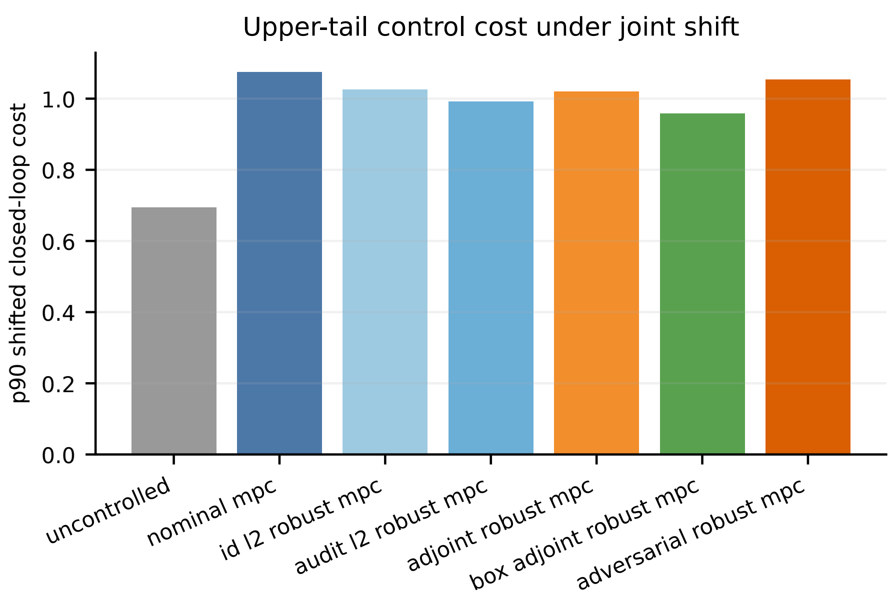

Figure 4: Empirical $p90$ closed-loop cost for the matched combined-shift actuator-gain sweep. The simultaneous-box adjoint controller gives the largest tail reduction, while the nonlinear adversarial controller is not best in this finite compute budget. Its row uses one zero start and three projected gradient-ascent steps, so it is not a globally optimized min–max baseline.

4.7 Navier–Stokes stress test: validity survives, localization weakens

On the official 128×128 test set, the base FNO has mean field RMSE 0.1635, where each sample uses the square root of its mean coordinate-wise squared error. This quantity is not relative L^2 error. The max-type multiplier is $q_{0.90} = 30.679$; simultaneous test coverage is 0.883 and mean full coordinate-band width is 4.3409. The pointwise disagreement scale correlates only $r = 0.169$ with absolute error. The realized count is 265/300, with descriptive 95%

Wilson interval $[0.842, 0.915]$, which contains the 0.90 target. Thus the observed 0.883 does not by itself diagnose systematic undercoverage at this sample size.

The multiplier and width reveal the more important limitation. A max score over 16,384 coordinates amplifies the largest standardized residual, while $r = 0.169$ shows that the local scale ranks pointwise errors weakly. Calibration then obtains approximate simultaneous validity mainly through a large global multiplier rather than accurate spatial adaptation. This negative result distinguishes a valid conformal wrapper from an informative uncertainty model: calibration can correct the event frequency, but it cannot create localization that is absent from the underlying scale.

Table 7: Independent-test Navier–Stokes field-UQ results. No action channel is available, so no control metric is reported.

Field RMSE	$q_{0.90}$	Simultaneous coverage	Full width	Scale–error r
0.1635	30.679	0.883	4.3409	0.169

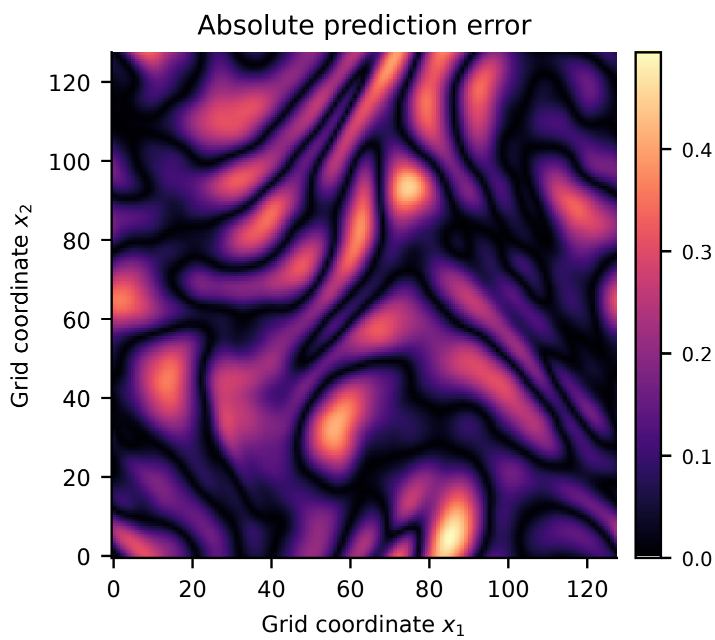

Figure 5: Representative absolute error of the two-dimensional FNO prediction. Errors are spatially structured rather than uniform, motivating a local scale; the following quantitative association test shows that the learned scale only weakly tracks them.

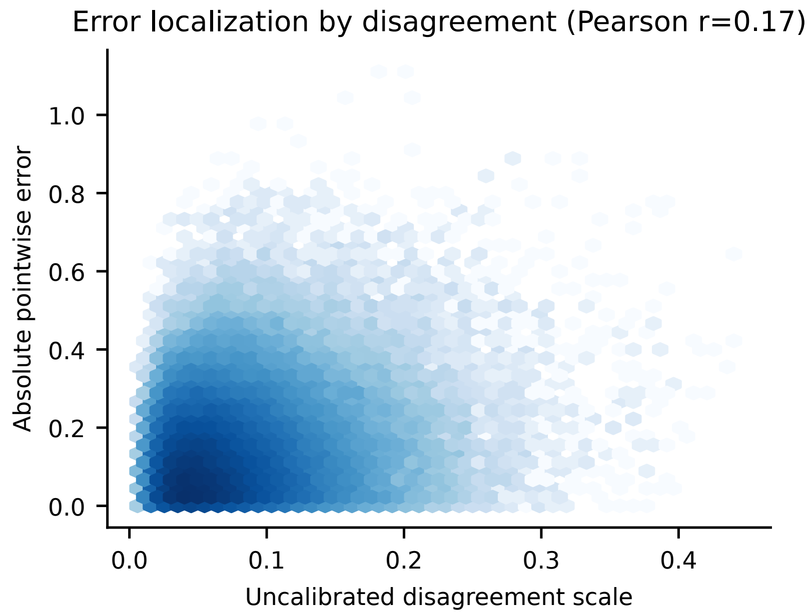

Figure 6: Pointwise perturbation scale versus absolute error on the independent Navier–Stokes test split. Pearson $r = 0.169$ reveals weak localization and helps explain the large simultaneous conformal multiplier.

4.8 Ablation C: tighter value bounds require matched coverage

Four raw bound constructions are scaled on the same 60 calibration trajectories to the 90% value-level target and evaluated on 160 independent trajectories at horizon 20 and $\gamma = 0.95$. The local recursion reaches 0.944 coverage with mean bound 0.0870. It is 11.9% narrower than the global recursion while attaining slightly higher empirical coverage. This is the only comparison that supports a clear reduction in conservatism at matched validity.

Adjoint support is wider, with mean bound 0.2035, and has lower coverage of 0.925. The fitted curvature coefficient is zero, so adding the curvature feature leaves the last two rows identical. Utilization above one corresponds to an uncovered test trajectory and cannot be read as tightness. The result shows that an adjoint direction can be useful for choosing robust actions without yielding the tightest scalar certificate of the realized value gap.

Table 8: Independent-test bound comparison. Utilization is observed value gap divided by the reported bound. Coverage must be read together with width.

Method	Coverage	Mean bound	Median η	$p90$ η	Max η
Global maximum	0.938	0.0987	0.210	0.688	3.919
Local recursion	0.944	0.0870	0.322	0.788	2.190
Adjoint support	0.925	0.2035	0.265	0.854	1.766
Adjoint + curvature	0.925	0.2035	0.265	0.854	1.766

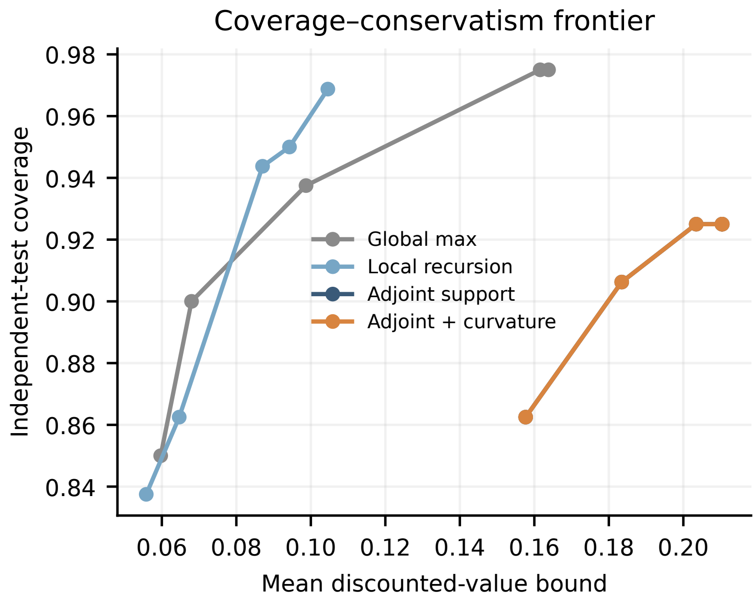

Figure 7: Coverage–mean-bound frontier on independent trajectories. A bound is less conservative only when it is narrower at matched coverage; utilization alone can be inflated by undercoverage.

4.9 Interpretation and limitations

What the current evidence supports. Three separations organize the evidence. Calibration population changes validity under compound shift; geometry changes support along the MPC adjoint; and the inner query changes how that support enters an action. In the controlled actuator sweep, the distinction is largest in the upper tail. The simultaneous box is wide in field space yet gives the lowest empirical $p90$ cost. Predictive width and decision utility are therefore not the same ordering.

What the current evidence does not support. The Burgers control result is one seed and one matched actuator sweep, not a multi-seed population claim. The public Navier–Stokes pairs have no action channel. The ordinary split-conformal theorem requires exchangeability with the deployment audit and does not cover a later distribution change. Marginal one-step coverage does not certify the stagewise uncertainty model queried by MPC. The behavior-trajectory result also does not cover counterfactual optimized actions. The deterministic value theorem assumes a uniform error and common closed-loop Lipschitz constants that were not certified for the learned FNO. The adjoint robust counterpart is first order unless a uniform curvature constant is established. At $H = 8$ and $n = 64$, the nonlinear inner problem has 512 perturbation coordinates for each candidate action sequence. Its state-dependent FNO rollout is nonconvex. Single-start three-step PGD supplies only a feasible lower bound on the worst-case cost, with no convergence test, multi-start check, or dual upper bound. The returned action therefore has no certificate against all perturbations in the stagewise set. Finite CEM adds a second optimization error that is also measured empirically rather than bounded.

Negative results guide the next experiment. The nonlinear adversarial controller is not best; the adjoint scalar bound is not tighter than the local recursion; the curvature feature adds nothing; and the Navier–Stokes perturbation scale has weak error association. These outcomes argue for the following next tests: multiple training/control seeds; audit-size and shift-severity sweeps; learned low-rank covariance or spectral function-space sets; policy-coupled trajectory calibration; and a controlled two-dimensional PDE with an explicit action channel. The adversarial audit also requires iteration and step-size sweeps, random restarts, and reporting the best feasible objective against the adjoint approximation. Deep ensembles, probabilistic neural operators, and learned quantile operators should be added as scale baselines under a matched total label and compute budget [12], probabilistic neural operators [5], and calibrated quantile operators [15].

5 Conclusion

Predictive coverage is not yet a control interface. This work makes that interface explicit by calibrating a field-valued dynamics set and querying its geometry along directions that change a finite-horizon objective. Exact support functions and adversarial rollouts provide two realizations, while a guarantee ledger keeps one-step conformal validity, deterministic value propagation, and empirical closed-loop performance separate.

The decisive controlled-Burgers comparison observes identical realized field coverage of 0.906 while changing geometry: the ellipsoid and box lead to empirical $p90$ costs of 1.0201 and 0.9586. This is conditional evidence from one matched actuator sweep, not a population or safety guarantee. The Navier–Stokes stress test exposes the complementary limitation, since high-dimensional validity can require wide sets when the local scale is weak. The next method should be held to a harder standard: policy-level trajectory coverage together with lower closed-loop cost at matched coverage and compute.

References

[1] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[2] Kavosh Asadi, Dipendra Misra, and Michael Littman. Lipschitz continuity in model-based reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 264–273, 2018.

[3] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.

[4] Nicolas Boullé and Alex Townsend. A mathematical guide to operator learning. In *Handbook of Numerical Analysis*, volume 25, pages 83–125. Elsevier, 2024.

[5] Christoph Bülte, Philipp Scholl, and Gitta Kutyniok. Probabilistic neural operators for functional uncertainty quantification. *Transactions on Machine Learning Research*, 2025. arXiv:2502.12902.

[6] Kong Yao Chee, Thales C. Silva, M. Ani Hsieh, and George J. Pappas. Uncertainty quantification and robustification of model-based controllers using conformal prediction. In *Proceedings of the 6th Annual Learning for Dynamics and Control Conference*, volume 242 of *Proceedings of Machine Learning Research*, pages 528–540, 2024.

[7] Jose Leopoldo Contreras, Ola Shorinwa, and Mac Schwager. Safe, out-of-distribution-adaptive mpc with conformalized neural network ensembles. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference*, volume 283 of *Proceedings of Machine Learning Research*, pages 194–207, 2025.

[8] Anushri Dixit, Lars Lindemann, Skylar X. Wei, Matthew Cleaveland, George J. Pappas, and Joel W. Burdick. Adaptive conformal prediction for motion planning among dynamic agents. In *Proceedings of the 5th Annual Learning for Dynamics and Control Conference*, volume 211 of *Proceedings of Machine Learning Research*, pages 300–314, 2023.

[9] Amir-massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1486–1494, 2017.

[10] Chancellor Johnstone and Bruce Cox. Conformal uncertainty sets for robust optimization. In *Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications*, volume 152 of *Proceedings of Machine Learning Research*, pages 72–90, 2021.

[11] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[12] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, pages 6402–6413, 2017.

[13] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[14] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

[15] Ziqi Ma, Danielle Pitt, Kamyar Azizzadenesheli, and Anima Anandkumar. Calibrated uncertainty quantification for operator learning via conformal prediction. *Transactions on Machine Learning Research*, 2024.

[16] Emilia Magnani, Nicholas Krämer, Runa Eschenhagen, Lorenzo Rosasco, and Philipp Hennig. Approximate bayesian neural operators: Uncertainty quantification for parametric PDEs. *Transactions on Machine Learning Research*, 2025. arXiv:2208.01565.

[17] Emilia Magnani, Marvin Pförtner, Tobias Weber, and Philipp Hennig. Linearization turns neural operators into function-valued gaussian processes. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025.

[18] Ali Malik, Volodymyr Kuleshov, Jiaming Song, Danny Nemer, Harlan Seymour, and Stefano Ermon. Calibrated model-based deep reinforcement learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 4314–4323, 2019.

[19] David Millard, Lars Lindemann, and Ali Baheri. Split conformal prediction in the function space with neural operators, 2025.

[20] S. Chandra Mouli, Danielle C. Maddix, Shima Alizadeh, Gaurav Gupta, Andrew Stuart, Michael W. Mahoney, and Bernie Wang. Using uncertainty quantification to characterize and improve out-of-domain learning for pdes. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 36372–36418, 2024.

[21] Christian Moya, Ali Mollaali, Zhen Zhang, Lu Lu, and Guang Lin. Conformalized-DeepONet: A distribution-free framework for uncertainty quantification in deep operator networks. *Physica D: Nonlinear Phenomena*, 471:134418, 2025.

[22] NeuralOperator Team. Navier–stokes dataset, 2024.

[23] Anutam Srinivasan, Antoine Leeman, and Glen Chou. Safety beyond the training data: Robust out-of-distribution mpc via conformalized system level synthesis. In *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pages 412–439, 2026.

[24] Shyam Sundhar Ramesh, Pier Giuseppe Sessa, Yifan Hu, Andreas Krause, and Ilija Bogunovic. Distributionally robust model-based reinforcement learning with large state spaces. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 100–108, 2024.

[25] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[26] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005.

[27] Weinan Wang, Bowen Gang, and Hao Deng. Operator learning for the 2d incompressible Navier–Stokes equations: A conformal prediction approach in the data-scarce regime, 2026.